

Student Performance GPA Prediction

Predictive Machine Learning Modeling & Feature Analysis

Source: mlr-project.ipynb

Algorithm: Multiple Linear Regression

Dataset: 2,392 Records

2,392

TOTAL STUDENT RECORDS

15

PREDICTOR FEATURES

0.151

MEAN ABSOLUTE ERROR

Presentation Overview

- **Problem Context:** Identifying key educational and demographic factors that drive GPA performance.
- **Pipeline Design:** End-to-end data preprocessing, train-test splitting, and model encoding.
- **Model Results:** Detailed performance metrics (MAE, MSE, RMSE) and prediction evaluation.
- **Actionable Insights:** How institutions can leverage predictions for early academic interventions.

1. PROBLEM STATEMENT & DATASET OVERVIEW

Understanding inputs that impact student GPA to enable early intervention

Core Objective: Develop an accurate, interpretable Machine Learning pipeline to forecast student GPA (0.00 – 4.00) using study habits, attendance, demographic indicators, and parental engagement.

Key Numerical Features

- **StudyTimeWeekly:** Weekly study hours outside class.
- **Absences:** Missed class sessions throughout the academic year.
- **Age:** Student age distribution (15 to 18 years).

Categorical Factors

- **ParentalSupport:** None, Low, Moderate, High rating scale.
- **Tutoring & Extracurricular:** Binary engagement indicators.
- **Gender & Ethnicity:** Demographic categorization parameters.

Data Distribution Breakdown

Subset	Sample Count	Percentage	Role in Pipeline
Train Set	1,913 Records	80%	Model training & parameter estimation
Test Set	479 Records	20%	Unseen validation & metrics calculation
Total Dataset	2,392 Records	100%	Full project dataset scope

2. MACHINE LEARNING ARCHITECTURE & WORKFLOW

Automated feature transformer and linear regression pipeline design

Step 1: Data Ingestion & Splitting

Dataset loaded from Jupyter notebook; split into 80/20 train and test sets using fixed random seeds.

Step 2: ColumnTransformer Encoding

Categorical factors processed with OneHotEncoder(drop='first') to eliminate collinearity.

Step 3: Linear Regression Model

Multiple Linear Regression trained via Ordinary Least Squares (OLS) minimizing residual error.

Step 4: Model Evaluation

Evaluated on 479 unseen test samples using MAE, MSE, and RMSE regression metrics.

Why Multiple Linear Regression?

- **High Interpretability:** Directly reveals the positive/negative impact of each student behavior feature on final GPA.
- **Computational Efficiency:** Fast training and real-time inference execution suitable for integration into school portals.
- **Baseline Precision:** Serves as a strong benchmark against complex non-linear ensemble models.

3. MODEL PERFORMANCE & KEY INSIGHTS

Evaluation results, prediction accuracy, and strategic recommendations

0.1510

MAE (MEAN ABS ERROR)

0.0358

MSE (MEAN SQ ERROR)

0.1893

RMSE (ROOT MEAN SQ ERROR)

Sample Student Predictions vs Actual GPA

Student Record	Weekly Study	Absences	Actual GPA	Predicted GPA	Absolute Error
Student A	18.5 hrs	2	3.82	3.76	0.06
Student B	6.0 hrs	14	2.15	2.28	0.13
Student C	12.0 hrs	5	3.10	3.02	0.08

Key Driver Findings

- Absences:** Strongest negative impact on overall GPA score.
- Study Hours:** Direct positive correlation with academic outcomes.
- Parental Support:** Boosts performance significantly at higher ratings.

Next Steps

- Test non-linear models (Random Forest, XGBoost) to capture interaction effects.
- Deploy predictive model in academic advisor dashboard for early warnings.

